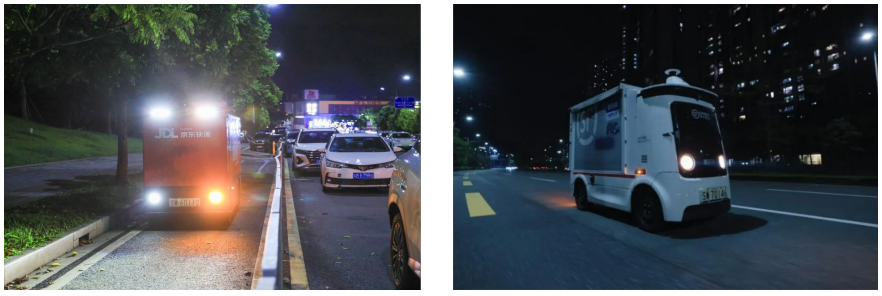
Stochastic Spatiotemporal Right-of-Way Allocation for Nighttime Autonomous Freight Transportation

1. Introduction

Nighttime deliveries offer a way to use spare road capacity and prepare goods for next-day service. In China, nighttime parcel collection and delivery services had expanded to nearly 300 cities by July 2024 (People's Daily, 2024). Moving freight at night can reduce competition with daytime passenger traffic and make better use of existing road capacity. These benefits, however, depend on drivers working overnight, when fatigue and irregular working hours pose challenges to safe and reliable operations.

Autonomous driving opens new possibilities for nighttime freight. Shenzhen authorized nighttime access for autonomous delivery vehicles (ADVs, as shown in Figure 1) in March 2026, and an August report recorded 437 nighttime routes, up from two initially (Shenzhen Longgang District Government, 2026). Driverless operation can reduce overnight driving labor, while coordinated platooning can save energy and shorter headways can increase effective capacity. Supporting these operations through reserved road space raises a planning question. Autonomous operation requires suitable road conditions throughout a trip, not just on selected segments. Where deployment relies on separated or strictly access-controlled road space, the availability of this space must be coordinated with vehicle routes and travel times. Deploying autonomous vehicles is therefore only part of the task; the road network must also provide the required access at the right locations and times.



a. JD ADV on dedicated lane **b. SF ADV on the road**

Figure 1: Examples of autonomous delivery vehicles operating on urban roads during nighttime periods.

In this paper, we study the temporary allocation of existing road capacity to autonomous freight during nighttime hours. An autonomous freight right-of-way (ROW) is a lane or portion of road capacity reserved exclusively for autonomous freight vehicles through physical separation or strict access control while active. A planner chooses which links to designate as ROWs and when to activate them, subject to a deployment-resource budget. Demand and

the road capacity remaining after background traffic are uncertain when these decisions are made. Once they are observed, a dispatcher assigns freight to autonomous transportation, conventional transportation, or an external service, and determines its departure times and routes. The objective is to minimize deployment costs and expected freight transportation costs.

The usefulness of a ROW depends on the other ROWs with which it connects. Consider two successive links on a route. If the first is open from midnight to 2 a.m. and the second from 2 a.m. to 4 a.m., a vehicle that completes the first link at 1:30 a.m. cannot continue immediately. A later departure may resolve the mismatch, but only if the shipment can still meet its delivery deadline. Moreover, a vehicle entering a link shortly before its ROW closes must be able to leave before access ends. Spatial connectivity alone is thus insufficient: the selected ROWs must support a time-feasible path and remain active throughout every link traversal.

Two sources of uncertainty complicate these decisions. Uncertain freight demand changes the volume of goods requiring transportation across locations and time periods, so ROW paths that are useful under one realization may be poorly matched to another. Meanwhile, fluctuations in background traffic change the remaining road capacity, affecting how much freight can be accommodated and which routes are attractive. Reserving capacity for autonomous operations may further constrain conventional freight on links with limited spare capacity. ROW schedules must therefore be chosen before either uncertainty is resolved, accounting for the routing and mode-allocation adjustments available afterward.

This paper studies the stochastic autonomous freight ROW allocation problem (SAFROWAP) and makes three contributions. First, we introduce a planning problem for the temporary allocation of road space to nighttime autonomous freight under uncertain demand and remaining capacity. The planner commits to a road-access schedule while retaining operational flexibility in shipment departures, routes, and transportation modes. The problem captures how a limited set of opening intervals can jointly support shipments with different origins, destinations, and delivery windows, and how the resulting transportation savings depend on the spatial and temporal arrangement of those intervals.

Second, we develop a two-stage stochastic formulation and an exact decomposition framework. Coarse design blocks represent road-access commitments, and a finer operational grid represents freight movements. Linking each autonomous traversal to every block it intersects enforces access throughout the journey while allowing different shipments to share the same opening decisions. The formulation retains continuous linear recourse and supports scenario-wise multi-cut Benders decomposition. Three network-reduction procedures, Pareto-oriented optimality cuts, and commodity-separable network-relaxation cuts address the size of the operational networks and strengthen the master problem's approximation of recourse costs.

Third, we design computational experiments to evaluate both the solution framework and the operational value of temporary ROW allocation. Adaptations of the R04–R10 benchmarks assess computational performance and the contribution of the algorithmic components. A Sioux Falls case study examines how deployment resources, opening-block duration, shipment timing, and uncertainty affect transportation costs and ROW utilization. The analysis investigates when concentrating resources on a few coordinated routes is effective and when broader spatial or temporal coverage provides greater flexibility, informing

the choice of where and when to reserve road space.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the stochastic autonomous freight ROW allocation problem. Section 4 presents the two-stage stochastic optimization formulation and its structural properties. Section 5 develops the decomposition algorithm and network-reduction procedures. Section 6 reports the computational design and performance measures.

2. Literature Review

This study lies at the intersection of autonomous freight infrastructure planning, service network design and stochastic network design. We review these three streams of literature and then position the present study with respect to their modeling and algorithmic foundation.

2.1. Autonomous Freight Transportation and Dedicated Infrastructure Design

The development of automated and connected vehicle technologies has stimulated a substantial literature on how road infrastructure should be adapted to accommodate automated traffic. Early studies mainly focused on the spatial deployment of dedicated infrastructure. Chen et al. (2016) optimize when, where, and how many autonomous-vehicle (AV) lanes should be deployed as AV market penetration evolves, while Chen et al. (2017) study the design of AV-only zones in a mixed-traffic network. Related work examines lane management and dedicated-lane deployment by explicitly accounting for the capacity effects of mixed human-driven and automated traffic (Ghiasi et al., 2017; Liu and Song, 2019). More recently, Li et al. (2020) study right-of-way (ROW) reallocation between AVs and human-driven vehicles, and Chakraborty et al. (2021) integrate AV-exclusive lane design with system-optimal dynamic traffic assignment. These studies establish that road space can be reallocated to exploit the operational advantages of automated vehicles, but their infrastructure decisions are generally spatial or strategic rather than an operational schedule that repeatedly activates and deactivates links within a day.

A growing set of studies has moved toward dynamic or uncertainty-aware lane management. Tani et al. (2022) consider dedicated CAV-lane deployment under stochastic traffic demand and stochastic lane capacity, and Zhang et al. (2023) develop a robust infrastructure deployment model that accounts for uncertain road capacity in mixed-autonomy traffic. Ngoduy et al. (2024) jointly characterize the spatial and temporal aspects of dedicated CAV-lane deployment using dynamic traffic assignment, while Jiang et al. (2025) optimize flexible lane allocation for dedicated automated-vehicle zones under time-varying traffic patterns. These papers are important methodological precedents for time-varying road-space management. Their primary focus, however, is mixed passenger traffic, traffic equilibrium, or network throughput rather than the design of a freight-specific access schedule that must support origin-to-destination freight movements under uncertain operational conditions.

The literature specifically addressing automated freight is more closely related to our setting. Fang et al. (2013) study an automated-truck transportation problem in which selected general-purpose lanes are reserved for automated trucks to provide time-efficient freight paths. She and Ouyang (2022) optimize the placement of dedicated truck-platooning

lanes and tolls while accounting for shipper route choice, fuel savings, pavement deterioration, and congestion. Xu et al. (2024) consider a lane-reservation-based autonomous-truck network design problem with carbon-emission objectives. A particularly relevant study is Zeng et al. (2024), which develops time-varying dedicated-lane planning for autonomous trucks under hours-of-service regulations. Most recently, Bhoopalam et al. (2026) formulate a scenario-based platoon-lane network design problem in which lane locations are selected before uncertain truck demand is realized and trucks are routed on a time-expanded network afterward. This work is especially close to the stochastic-planning aspect of our problem, but its first-stage design selects a permanent set of platooning arcs; temporal activation of dedicated platooning lanes is explicitly left as a future extension.

Another closely related stream concerns time-dependent access regulation. Esfandeh et al. (2018) determine when selected road sections should be temporarily closed to hazardous-material vehicles, anticipating the carriers' subsequent route and departure-time choices. Their model shows that road accessibility itself can be designed at a within-day temporal resolution and that link availability must be consistent with the entire traversal of a vehicle. Nevertheless, the problem is deterministic and regulatory: the network administrator restricts access to reduce hazardous-material risk, while independent carriers choose least-cost route-time alternatives. Our setting instead uses link-time decisions to activate an enhanced ROW layer for autonomous freight, and these decisions must be made before freight demand and background traffic conditions are known. Thus, the closest existing studies separately capture either temporal access design or stochastic autonomous-freight infrastructure design, but not their combination.

2.2. Service Network Design

A second stream relevant to our work is service network design (SND), which provides a well-established framework for tactical freight planning. In the classic formulation, SND coordinates the selection and scheduling of transportation services, terminal operations, and freight routing to exploit consolidation economies (Crainic, 2000). Large-scale LTL applications were developed by Jarrah et al. (2009). Scheduled service network design (SSND) makes the temporal dimension explicit by coordinating service schedules and shipment movements. For example, Zhu et al. (2014) integrate service selection and scheduling, train makeup, blocking, and shipment routing in a cyclic space-time network for rail freight. Boland et al. (2017) develop a continuous-time SND formulation and a dynamic discretization approach, and Hewitt (2022) extend SSND by allowing selected shipments to flex their available and due times to create additional consolidation opportunities.

The emergence of automated driving has also entered the SND literature. Scherr et al. (2019) jointly determine fleet mix and the routing of vehicles and goods in a heterogeneous infrastructure with manually driven and autonomous vehicles. Scherr et al. (2020) further develop an exact dynamic discretization approach for this mixed-autonomous-fleet SND problem, where feasible streets for autonomous operation are part of the exogenous infrastructure. Al Hajj Hassan et al. (2022) examine daily load planning under alternative autonomous-truck deployment scenarios, showing how automation can alter LTL load-plan decisions. Scherr et al. (2022) consider a stochastic SND problem for a platooning service provider, where first-stage decisions determine fleet size and scheduled platooning services and second-stage decisions accept requests and route AVs through those services.

These studies are closely related to our second-stage problem because both require freight movements to be feasible in space and time. The distinction lies in the object being designed. In SND and SSND, a design decision creates or schedules a carrier-operated transportation service, such as a vehicle dispatch or service capacity, and shipment consolidation is a central economic mechanism. Our model does not determine vehicle dispatches or transportation services. Instead, it determines the time-dependent accessibility of road infrastructure; after the ROW schedule is fixed and uncertainty is realized, freight is routed over the resulting multilayer space-time network. Hence, the SND literature provides an important foundation for representing end-to-end temporal feasibility, but addresses a fundamentally different design object.

2.3. Stochastic Network Design

A third relevant stream concerns network design under uncertainty. In freight SND, Lium et al. (2009) demonstrate the value of explicitly accounting for stochastic demand when constructing service networks. Bai et al. (2014) introduce second-stage vehicle rerouting to increase flexibility after demand is realized. Wang et al. (2019) investigate the value and upgradability of deterministic solutions for stochastic scheduled transportation network design. Hewitt et al. (2019) jointly consider resource acquisition, service-network planning, and uncertain freight volumes, while Baubaid et al. (2021) quantify the value of limited routing flexibility in two-stage stochastic LTL service-network designs. Complementing demand uncertainty, Lanza et al. (2021) formulate stochastic SSND with uncertain travel times and delivery-quality targets, and Liu et al. (2023) study a two-stage stochastic SND in which some vehicle routes are fixed before demand is observed and others can adapt afterward.

The broader stochastic multicommodity network design literature is particularly relevant methodologically. Crainic et al. (2021) study a two-stage stochastic capacitated multicommodity network design problem in which both demands and arc capacities are uncertain and develop a partial Benders decomposition that retains selected scenario information in the master problem. Sarayloo et al. (2021) develop a learning-based matheuristic for stochastic capacitated multicommodity network design. In a city-logistics setting, Nieto-Isaza et al. (2022) consider both stochastic demand and stochastic time-dependent transportation capacity, with first-stage infrastructure decisions and second-stage space-time commodity allocation, and propose an enhanced Benders decomposition. These studies establish the standard information structure that is also central to our work: infrastructure or network decisions are made before uncertainty is observed, and flow decisions provide recourse afterward.

Uncertainty has also been considered in AV infrastructure planning, but with a different focus. Chen et al. (2019) formulate a two-stage stochastic AV incentive-design problem in which AV-lane deployment is decided before uncertain vehicle purchase prices are observed, whereas second-stage subsidies adapt to the realized market condition. Tani et al. (2022) account for stochastic traffic demand and capacity when deploying dedicated CAV lanes, and Zhang et al. (2023) focus on robust infrastructure deployment under capacity uncertainty. Most closely, Bhoopalam et al. (2026) design dedicated truck platoon lanes under scenario-based uncertain truck demand and use scenario-specific time-expanded routing as recourse. These studies show that uncertainty is already relevant to automated-vehicle infrastructure planning. However, their design variables describe strategic lane locations or permanent infrastructure configurations rather than an explicit within-day link-time access schedule.

Positioning of this study.. The literature therefore provides three complementary foundations. Autonomous-vehicle infrastructure studies explain how dedicated road space can improve automated traffic operations; SND and SSND provide detailed representations of freight movements and their end-to-end temporal feasibility; and stochastic network design provides the design-before-uncertainty and routing-after-uncertainty framework. Yet the intersection of these three streams remains limited. The closest temporal-access model, Esfandeh et al. (2018), is deterministic and restrictive; the time-varying autonomous-truck lane model of Zeng et al. (2024) does not explicitly model operational uncertainty; and the stochastic platoon-lane design of Bhoopalam et al. (2026) assumes permanent lane locations and explicitly points to temporal network designs as a future direction. Building on these streams, we consider a two-stage stochastic, within-day ROW design problem in which first-stage link-time decisions activate an enhanced autonomous-freight layer before freight demand and background traffic conditions are realized. Scenario-specific recourse then routes freight over the resulting multilayer space-time network while respecting end-to-end time feasibility and available capacity. This combination shifts the design object from permanent infrastructure or scheduled transportation services to *time-dependent infrastructure access under uncertainty.*

3. Problem Description

3.1. Operating setting

We consider a road authority that allocates existing road capacity to autonomous freight during a nighttime planning horizon. An autonomous freight ROW is a lane or portion of road capacity reserved exclusively for ADVs through physical separation or strict access control. While active, the reserved capacity is unavailable to conventional freight vehicles and passenger vehicles.

We divide the nighttime planning horizon into consecutive time intervals, called *coarse design blocks*, within which the ROW status of each link remains unchanged. For example, a six-hour horizon may contain six one-hour design blocks. Each block is further divided into shorter *operational periods*, which represent freight movements and road capacities. Thus, a one-hour design block contains six operational periods when the finer interval is 10 minutes.

These two time scales support a two-stage decision process. Before freight demand and background traffic volumes are known, the authority decides which links to reserve in each design block, subject to an aggregate deployment-resource budget. Background traffic occupies general-road capacity, leaving a scenario-dependent amount available for the freight considered in the model. After demand and background traffic are observed, a dispatcher determines freight allocation, departure times, and routes on the finer operational grid. Each autonomous traversal must be covered by active ROW blocks throughout its duration; a traversal extending across a block boundary requires both blocks to be active on that link.

Each freight commodity is defined by an origin, a destination, a release time, and a delivery deadline, with its quantity varying across scenarios. Demand is divisible and may be allocated among autonomous transportation, conventional trucking, and an external transportation option. Freight assigned to either road-based mode may remain at its origin terminal before departure, but must then travel to its destination without intermediate waiting

or a change of mode. Autonomous freight requires a complete ROW path within its delivery window. Conventional freight uses the general road network, subject to the capacity remaining after background traffic and ROW reservation.

The objective is to minimize ROW deployment costs and expected freight transportation costs, including external transportation. Autonomous operation offers potential labor and energy savings, as well as travel-time and throughput advantages. Reserving road space, however, reduces the capacity available to conventional freight. The planner therefore selects opening locations and times jointly, accounting for the autonomous routes they enable and the cost of serving the remaining freight.

3.2. Modeling assumptions

The formulation is developed under the following assumptions.

Assumption 1. *Freight movements are centrally coordinated and represented as divisible aggregate flows.*

Assumption 2. *A shipment may be held at its origin logistics terminal before dispatch. After dispatch, it travels continuously to its destination without waiting at intermediate road nodes. An autonomous shipment must remain in the autonomous layer and use active ROWs throughout every link traversal.*

Assumption 3. *Nighttime passenger traffic is exogenous, and its effect on freight transportation is represented by the scenario-dependent residual capacities of general-purpose roads.*

Assumption 4. *Freight that cannot be accommodated by either transportation mode may be assigned to an emergency service at a sufficiently high unit cost.*

Assumption 1 represents centrally coordinated freight operations at an aggregate level. Each commodity comprises multiple shipments, with demand measured in truckload equivalents and divisible among routes, departure times, and transportation modes. Assumption 2 permits holding at origin terminals but excludes discretionary waiting at intermediate road nodes. Autonomous freight must follow an end-to-end ROW path, with access maintained throughout every link traversal; otherwise, it must use conventional transportation or the external service. Assumption 3 treats background traffic as exogenous. Its effects are represented through scenario-dependent residual road capacities, allowing the model to capture capacity uncertainty without explicitly modeling background traffic routing. Assumption 4 provides an external service at a high but finite unit cost. Together with nonnegative residual capacities under every admissible ROW schedule, its unlimited availability ensures relatively complete recourse. External service use measures demand served outside the planned road system, whether because of capacity shortages or costly routing alternatives.

3.3. Construction of the Two-Layer Time-Expanded Network

We represent freight movements on two layers indexed by $\mathcal{M} = \{A, H\}$, corresponding to autonomous and conventional transportation. Both layers use the same fine operational grid, while ROW decisions are defined over coarser design blocks.

Let $G = (\mathcal{N}, \mathcal{L})$ be the directed road network, and let $\mathcal{L}^A \subseteq \mathcal{L}$ contain the links eligible for ROW operation. The operational grid has timestamps $\mathcal{T} = \{0, \ldots, H\}$, separated by δ units of time. Operational period $t \in \{0, \ldots, H-1\}$ spans timestamps t and $t+1$.

Let $\mathcal{P}$ be the index set of consecutive design blocks, and let $\mathcal{T}_p$ contain the operational periods in block $p \in \mathcal{P}$. These sets partition $\{0, \ldots, H-1\}$. The mapping $\kappa(t) = p$ identifies the block containing operational period t. For equal block duration L_D, each block contains $|\mathcal{T}_p| = L_D/\delta$ operational periods, assumed integer.

The candidate set $\mathcal{C} \subseteq \mathcal{L}^A \times \mathcal{P}$ contains the link–block combinations eligible for ROW activation. With these candidates and block boundaries fixed, refining the operational grid does not increase the number of first-stage design variables.

Each commodity $k \in \mathcal{K}$ has an origin o_k, a destination q_k, a release time r_k, and a deadline h_k, where $r_k, h_k \in \mathcal{T}$ and $r_k \leq h_k$. Its demand d_k^ω varies by scenario. For each node i, timestamp t, and mode m, the node v_{it}^m represents freight located at i at time t in layer m.

Link travel times τ_ℓ^m are fixed positive integers measured in operational periods. We assume

$$\tau_\ell^A \leq \tau_\ell^H, \qquad \ell \in \mathcal{L}^A, \tag{1}$$

allowing equal discretized durations when rounding removes the autonomous travel-time advantage on short links.

The two layers contain travel, no en-route waiting arcs, origin-dispatch, and delivery arcs.

- **Travel arcs.** For link $\ell = (i, j)$ and timestamp $t \in \mathcal{T}$ satisfying $t + \tau_\ell^m \leq H$, a travel arc

$$e = \left(v_{it}^m, v_{j,t+\tau_\ell^m}^m\right) \tag{2}$$

represents freight entering the link at t and reaching its downstream node at $t + \tau_\ell^m$. Conventional travel arcs use links in $\mathcal{L}$. For an autonomous traversal, define its occupied operational periods and the design blocks it intersects as

$$\mathcal{I}_{\ell t}^A = \{t, t+1, \ldots, t+\tau_\ell^A - 1\}, \tag{3}$$

$$\mathcal{P}_{\ell t}^A = \{p \in \mathcal{P} : \mathcal{I}_{\ell t}^A \cap \mathcal{T}_p \neq \varnothing\}. \tag{4}$$

An autonomous travel arc is included only if $\ell \in \mathcal{L}^A$ and $(\ell, p) \in \mathcal{C}$ for every $p \in \mathcal{P}_{\ell t}^A$. Its flow is subsequently restricted by the activation decisions for all these blocks, ensuring ROW access throughout the traversal. Let $\mathcal{E}^{m,T}$ denote all candidate travel arcs in layer m. For each commodity k, let $\mathcal{E}_k^{m,T} \subseteq \mathcal{E}^{m,T}$ contain those lying on at least one path from o_k to q_k that departs no earlier than r_k and arrives no later than h_k. These paths are identified in the candidate network before ROW activation decisions are made.

- **No en-route waiting arcs.** Freight cannot wait at intermediate road nodes in either layer:

$$\mathcal{E}_k^{m,W} = \varnothing, \qquad k \in \mathcal{K}, \quad m \in \mathcal{M}. \tag{5}$$

Ordinary intersection delays are included in link travel times. No transfer arcs connect the two layers, so freight retains its assigned transportation mode throughout its trip.

- **Origin-dispatch arcs.** For each commodity k and layer m, introduce a virtual origin terminal σ_k^m. Let $D^m(o_k, q_k)$ denote the shortest-path duration computed using τ_ℓ^m, with links restricted to $\mathcal{L}^A$ for autonomous transportation. The dispatch-arc set is

$$\mathcal{E}_k^{m,R} = \left\{(\sigma_k^m, v_{o_k t}^m) : t \in \mathcal{T},\ r_k \leq t \leq h_k - D^m(o_k, q_k)\right\}, \qquad m \in \mathcal{M}. \tag{6}$$

These arcs allow freight to remain at its origin terminal before dispatch, with divisible demand assigned to different departure times. The shortest-path bound excludes departures that cannot meet the deadline even without capacity restrictions. If no physical path exists in a layer, its dispatch set is empty.

- **Delivery arcs.** For each commodity k and layer m, introduce a dummy sink d_k^m and the delivery-arc set

$$\mathcal{E}_k^{m,D} = \left\{(v_{q_k t}^m, d_k^m) : t \in \mathcal{T},\ r_k \le t \le h_k\right\}.$$

These arcs connect admissible arrival times to a common sink for the flow-balance constraints.

The resulting commodity network is $\mathcal{G}_k^m = (\mathcal{V}_k^m, \mathcal{E}_k^m)$, where

$$\mathcal{E}_k^m = \mathcal{E}_k^{m,R} \cup \mathcal{E}_k^{m,T} \cup \mathcal{E}_k^{m,D}, \qquad m \in \mathcal{M}. \tag{7}$$

Nodes and arcs outside every source-to-sink path satisfying its time window are removed; Section 5.2 describes the reduction procedures. For $v \in \mathcal{V}_k^m$, let $\delta_{km}^+(v)$ and $\delta_{km}^-(v)$ denote its outgoing and incoming arcs.

For an entry pair (ℓ, t), let $\mathcal{E}_{k,\ell t}^{m,T} \subseteq \mathcal{E}_k^{m,T}$ contain the arcs entering link ℓ at time t. With $f_{ke}^{m,\omega}$ denoting scenario freight flow, the aggregate autonomous entry flow is

$$F_{\ell t}^{A,\omega} = \sum_{k \in \mathcal{K}} \sum_{e \in \mathcal{E}_{k,\ell t}^{A,T}} f_{ke}^{A,\omega}. \tag{8}$$

We write $\mathcal{A}^m = \{(\ell, t) : \mathcal{E}_{k,\ell t}^{m,T} \neq \varnothing \text{ for some } k\}$ for the retained entry pairs. Capacities limit link entries per operational period, measured in truckload equivalents. Any clearance required for ROW activation is assumed to be completed before opening; vehicle-level lane transitions are not modeled.

This construction allows origin holding followed by travel without intermediate waiting or modal transfers. Under a binary ROW schedule, every autonomous traversal requires $y_{\ell p} = 1$ for all $p \in \mathcal{P}_{\ell t}^A$. A traversal extending across a block boundary therefore requires the intersected blocks to be active, whereas one ending exactly at the boundary does not require the following block. Freight without a feasible autonomous path must use conventional transportation, subject to capacity and time-window constraints, or the external service.

3.4. Notation

Table 1 summarizes the notation used in the formulation. Freight flows, demand, and entry capacities are measured in truckload equivalents. Time indices and travel durations are expressed in operational periods of length δ. Each scenario $\omega \in \Omega$ has probability $p_\omega > 0$, with $\sum_{\omega \in \Omega} p_\omega = 1$.

For each commodity network, the flow-balance coefficient is

$$b_{kv}^m = \begin{cases} 1, & v = \sigma_k^m, \\ -1, & v = d_k^m, \\ 0, & \text{otherwise.} \end{cases}$$

The first-stage variables $y_{\ell p}$ are binary. All second-stage flow, mode-allocation, and external-service variables are nonnegative and continuous.

Table 1: Main notation.

Symbol	Definition
Sets and network elements	
$\mathcal{N};\ \mathcal{L}$	Physical road nodes and directed links.
$\mathcal{L}^A$	Subset of links eligible for ROW operation.
$\mathcal{M} = \{A, H\}$	Autonomous and conventional transportation modes.
$\mathcal{K};\ \Omega$	Freight commodities and stochastic scenarios.
$\mathcal{T} = \{0, \dots, H\}$	Operational timestamps; H is the number of operational periods.
$\mathcal{P}$	Index set of coarse design blocks.
$\mathcal{T}_p$	Operational periods contained in design block p.
$\mathcal{C} \subseteq \mathcal{L}^A \times \mathcal{P}$	Candidate link–block combinations for ROW activation.
$\mathcal{G}_k^m = (\mathcal{V}_k^m, \mathcal{E}_k^m)$	Commodity network for k in mode m, with node set $\mathcal{V}_k^m$ and arc set $\mathcal{E}_k^m$.
v_{it}^m	Copy of physical node i at timestamp t in layer m.
$\sigma_k^m;\ d_k^m$	Virtual origin terminal and dummy destination sink of the commodity network.
$\mathcal{E}^{m,T}$	All candidate travel arcs in layer m.
$\mathcal{E}_k^{m,T}$	Travel arcs retained for commodity k in layer m.
$\mathcal{E}_k^{m,R};\ \mathcal{E}_k^{m,D}$	Origin-dispatch and delivery arcs for commodity k in layer m.
$\mathcal{E}_k^{m,W}$	Intermediate waiting-arc set, empty in both layers.
$\mathcal{E}_{k,\ell t}^{m,T}$	Travel arcs of commodity k entering link ℓ at time t in mode m.
$\mathcal{A}^m$	Link–entry-period pairs associated with retained travel arcs in layer m.
$\mathcal{I}_{\ell t}^A$	Operational periods occupied by an autonomous traversal entering link ℓ at time t.
$\mathcal{P}_{\ell t}^A$	Design blocks intersected by that autonomous traversal.
$\delta_{km}^+(v);\ \delta_{km}^-(v)$	Outgoing and incoming arcs of node v in $\mathcal{G}_k^m$.
Parameters and mappings	
$\delta;\ L_D$	Duration of an operational period and a design block, respectively.
$\kappa(t)$	Index of the design block containing operational period t.
$o_k;\ q_k$	Origin and destination of commodity k.
$r_k;\ h_k$	Release time and delivery deadline of commodity k.
τ_ℓ^m	Travel duration on link ℓ in mode m, measured in operational periods.
$D^m(i, j)$	Shortest-path duration from i to j, using τ_ℓ^m as link weights and restricting links to $\mathcal{L}^A$ when $m = A$.
p_ω	Probability of scenario ω.

Continued on next page

Table 1 (continued)

Symbol	Definition
d_k^ω	Realized demand of commodity k in scenario ω.
B	Total deployment-resource budget.
$\rho_{\ell p};\ g_{\ell p}$	Resource requirement and deployment cost of activating ROW cell (ℓ, p).
$U_{\ell t}^{A,\omega}$	Autonomous entry capacity on link ℓ in period t when all ROW blocks required for the traversal are active.
$R_{\ell t}^{H,\omega}$	General-road entry capacity remaining after background traffic, before ROW reservation.
$\Delta_{\ell t}$	General-road entry capacity removed in period t when ROW block $(\ell, \kappa(t))$ is activated.
$c_e^{m,\omega}$	Unit freight operating cost on arc e in mode m and scenario ω.
M_k	Unit external-service cost for commodity k.
b_{kv}^m	Flow-balance coefficient: 1 at the virtual source, -1 at the dummy sink, and 0 elsewhere.
Decision variables	
$y_{\ell p}$	Equal to 1 if link ℓ is reserved as a ROW throughout block p, and 0 otherwise.
$f_{ke}^{m,\omega}$	Flow of commodity k on arc e in mode m and scenario ω.
$z_k^{m,\omega}$	Amount of commodity k assigned to mode m in scenario ω.
s_k^ω	Amount of commodity k assigned to the external service in scenario ω.
Derived quantities	
$F_{\ell t}^{A,\omega}$	Aggregate autonomous freight entering link ℓ in period t, obtained by summing commodity flows.
$Q_\omega(y)$	Minimum second-stage operating cost in scenario ω under ROW schedule y.

4. Mathematical Formulation

We formulate the ROW allocation problem as a two-stage stochastic program. The first stage determines a common ROW schedule across scenarios. The second stage assigns and routes realized freight demand on the two-layer network constructed in Section 3.3.

4.1. First-stage ROW decisions

Let $y_{\ell p}$ equal one if link ℓ is reserved as a ROW throughout design block p, and zero otherwise. The feasible design set is

$$\mathcal{Y} = \left\{ y \in \{0,1\}^{|\mathcal{C}|} : \sum_{(\ell,p)\in\mathcal{C}} \rho_{\ell p} y_{\ell p} \le B \right\}, \tag{9}$$

where $\rho_{\ell p} > 0$ is the resource required to activate cell (ℓ, p) and $B \geq 0$ is the available budget. The resource may represent controlled lane-hours or deployment resources such as monitoring equipment. Each activation covers the entire design block; consecutive active blocks provide a longer opening.

Let $g_{\ell p} \geq 0$ be the deployment cost of cell (ℓ, p) and $Q_\omega(y)$ the minimum operating cost in scenario ω under schedule y. The first-stage problem is

$$\min_{y \in \mathcal{Y}} \left\{ \sum_{(\ell,p) \in \mathcal{C}} g_{\ell p} y_{\ell p} + \sum_{\omega \in \Omega} p_\omega Q_\omega(y) \right\}. \tag{10}$$

Deployment decisions are shared by all scenarios, while freight allocation and routing adapt to realized demand and available capacity.

4.2. Scenario recourse problem

For each commodity k and scenario ω, $z_k^{m,\omega}$ denotes the quantity assigned to mode m, $f_{ke}^{m,\omega}$ its flow on arc e, and s_k^ω the quantity assigned to the external service. All these variables are nonnegative and continuous. The aggregate autonomous entry flow $F_{\ell t}^{A,\omega}$ is defined in (8).

The capacity parameters are finite and nonnegative. For every entry pair controlled by a candidate ROW block, we assume

$$0 \leq \Delta_{\ell t} \leq R_{\ell t}^{H,\omega}, \qquad (\ell, t) \in \mathcal{A}^H : (\ell, \kappa(t)) \in \mathcal{C}, \quad \omega \in \Omega. \tag{11}$$

Thus, ROW reservation cannot reduce conventional entry capacity below zero. Demand and unit costs are finite and nonnegative, and external service is available without a capacity limit at unit cost M_k.

For fixed $y \in \mathcal{Y}$, the scenario problem is

$$Q_\omega(y) = \min_{f,z,s} \quad \sum_{k \in \mathcal{K}} \sum_{m \in \mathcal{M}} \sum_{e \in \mathcal{E}_k^m} c_e^{m,\omega} f_{ke}^{m,\omega} + \sum_{k \in \mathcal{K}} M_k s_k^\omega \tag{12}$$

$$\text{s.t.} \quad z_k^{A,\omega} + z_k^{H,\omega} + s_k^\omega = d_k^\omega, \qquad k \in \mathcal{K}, \tag{13}$$

$$\sum_{e \in \delta_{km}^+(v)} f_{ke}^{m,\omega} - \sum_{e \in \delta_{km}^-(v)} f_{ke}^{m,\omega} = b_{kv}^m z_k^{m,\omega}, \qquad \begin{array}{l} k \in \mathcal{K},\ m \in \mathcal{M}, \\ v \in \mathcal{V}_k^m, \end{array} \tag{14}$$

$$F_{\ell t}^{A,\omega} \leq U_{\ell t}^{A,\omega} y_{\ell p}, \qquad \begin{array}{l} (\ell, t) \in \mathcal{A}^A, \\ p \in \mathcal{P}_{\ell t}^A, \end{array} \tag{15}$$

$$\sum_{k \in \mathcal{K}} \sum_{e \in \mathcal{E}_{k,\ell t}^{H,T}} f_{ke}^{H,\omega} \leq R_{\ell t}^{H,\omega} - \Delta_{\ell t} y_{\ell,\kappa(t)}, \qquad \begin{array}{l} (\ell, t) \in \mathcal{A}^H, \\ (\ell, \kappa(t)) \in \mathcal{C}, \end{array} \tag{16}$$

$$\sum_{k \in \mathcal{K}} \sum_{e \in \mathcal{E}_{k,\ell t}^{H,T}} f_{ke}^{H,\omega} \leq R_{\ell t}^{H,\omega}, \qquad \begin{array}{l} (\ell, t) \in \mathcal{A}^H, \\ (\ell, \kappa(t)) \notin \mathcal{C}, \end{array} \tag{17}$$

$$f_{ke}^{m,\omega} \geq 0, \qquad k \in \mathcal{K},\ m \in \mathcal{M},\ e \in \mathcal{E}_k^m, \tag{18}$$

$$z_k^{m,\omega} \geq 0, \qquad s_k^\omega \geq 0, \qquad k \in \mathcal{K},\ m \in \mathcal{M}. \tag{19}$$

Constraint (13) allocates all realized demand among autonomous transportation, conventional transportation, and external service. Together with nonnegativity, it bounds each allocated quantity by d_k^ω. Constraint (14) routes the quantity assigned to each mode from its virtual origin terminal to its destination sink. Release times, delivery deadlines, and the absence of intermediate waiting and modal transfers are enforced by the network construction.

Constraint (15) links the operational and design grids. An autonomous entry flow is bounded by every ROW block intersected by its traversal. For binary y, positive flow therefore requires all these blocks to be active. A traversal may end exactly when a block closes; if it continues beyond that boundary, the following block must also be active.

Constraints (16) and (17) limit conventional entries. The parameter $R_{\ell t}^{H,\omega}$ represents capacity remaining after background traffic, while $\Delta_{\ell t}$ is the capacity removed by ROW reservation. Consequently, opening a ROW can enable autonomous transportation while restricting conventional routing.

The objective (12) includes operating costs in both layers and external-service costs. Travel-arc costs represent vehicle, labor, energy, and autonomous-operation expenses as applicable. Origin-dispatch costs can represent holding before departure, while delivery arcs may have zero cost. External service is priced to discourage its use, but may be selected when regular routes are unavailable or sufficiently costly.

4.3. Model properties

Let

$$\overline{\mathcal{Y}} = \left\{ y \in [0,1]^{|\mathcal{C}|} : \sum_{(\ell,p)\in\mathcal{C}} \rho_{\ell p} y_{\ell p} \le B \right\} \tag{20}$$

be the continuous relaxation of the design set. The recourse formulation also defines $Q_\omega(y)$ for $y \in \overline{\mathcal{Y}}$.

Proposition 1 (Recourse structure)**.** *For every $y \in \overline{\mathcal{Y}}$ and $\omega \in \Omega$, the recourse problem is feasible and has a finite optimal value satisfying*

$$0 \le Q_\omega(y) \le \sum_{k\in\mathcal{K}} M_k d_k^\omega. \tag{21}$$

Moreover, Q_ω is continuous, convex, and piecewise linear on $\overline{\mathcal{Y}}$.

Proof. Set all road flows and mode allocations to zero and let $s_k^\omega = d_k^\omega$ for every commodity. This solution satisfies the demand and flow-balance constraints. All capacity right-hand sides are nonnegative, including those for conventional traffic by (11). The solution is therefore feasible and has cost $\sum_k M_k d_k^\omega$. Nonnegative costs give the lower bound.

The recourse problem is a finite-dimensional linear program whose constraint matrix and objective coefficients are independent of y and whose right-hand side is affine in y. By linear programming duality, its finite value function is polyhedral and convex, and hence continuous and piecewise linear on $\overline{\mathcal{Y}}$. □

Proposition 1 establishes relatively complete recourse. Fixing the ROW schedule leaves independent linear programs across scenarios, allowing multi-cut Benders decomposition with optimality cuts only.

Proposition 2 (Path decomposition and ROW protection)**.** *For every $y \in \overline{\mathcal{Y}}$, each feasible commodity flow in either layer admits a decomposition into source-to-sink paths satisfying its release time and deadline, without intermediate waiting or modal transfers. If $y \in \mathcal{Y}$ is binary, every positive-flow autonomous path is protected by active ROW blocks throughout each link traversal.*

Proof. Every travel arc advances time because τ_ℓ^m is a positive integer. Virtual sources have only outgoing dispatch arcs, and dummy sinks have only incoming delivery arcs. Each commodity network is therefore acyclic. Flow conservation decomposes its feasible flow into nonnegative source-to-sink path flows.

Dispatch arcs enforce the release time, and delivery arcs enforce the deadline. Since there are no intermediate waiting arcs, a path continuing from an intermediate node must depart at its arrival timestamp. The absence of transfer arcs preserves its mode.

Now let y be binary. If an autonomous path carries positive flow on an arc entering link ℓ at t, then $F_{\ell t}^{A,\omega} > 0$. Constraint (15) requires $y_{\ell p} = 1$ for every $p \in \mathcal{P}_{\ell t}^A$. These blocks cover the entire traversal, establishing end-to-end ROW protection. □

The proposition shows that route continuity and ROW protection follow from the network structure and capacity constraints without route-level binary variables.

Remark 1 (Capacity transfer). If $\Delta_{\ell t} = 0$ for all entry periods controlled by candidate blocks, then $Q_\omega(y)$ is componentwise nonincreasing on $\overline{\mathcal{Y}}$. Increasing y relaxes autonomous capacity constraints and leaves conventional capacity unchanged. With positive capacity transfer, this monotonicity need not hold: an activation may reduce autonomous costs but increase the cost of serving conventional freight.

Remark 2 (Complementarity of ROW activations). Consider one unit of demand whose only feasible autonomous path requires two link–block cells, a and b. Suppose both can be activated within the budget, capacity transfer is zero, and autonomous service saves $\eta > 0$ relative to the best alternative. Activating either cell alone yields no operating benefit, whereas activating both yields η. The marginal benefit of activating b therefore increases when a is already active. Thus, operating-cost savings from ROW activation are not submodular in general, and the value of an activation cannot be assessed independently of the paths it helps complete.

5. Solution Approach

5.1. Decomposition structure

Fixing the ROW schedule leaves an independent linear recourse problem for each scenario. Proposition 1 establishes relatively complete recourse, so a multi-cut Benders method requires only optimality cuts.

The solution procedure first reduces the commodity networks and then combines two cut mechanisms. Pareto-oriented cuts select informative dual solutions from the exact scenario problems. Commodity-separable network-relaxation cuts provide initial lower bounds

by relaxing competition for shared capacity. These relaxation cuts are generated during initialization, while exact scenario cuts drive the main Benders loop.

5.2. Exact reduction of the time-expanded networks

Three preprocessing rules reduce the commodity networks. The first two identify temporally feasible paths; the third uses the ROW resource budget to exclude autonomous movements that no feasible binary design can support. The candidate set $\mathcal{C}$ and the first-stage feasible set $\mathcal{Y}$ remain unchanged.

R1: Shortest-time window screening.. A node v_{it}^m can lie on a feasible path for commodity k only if

$$r_k + D^m(o_k, i) \leq t \leq h_k - D^m(i, q_k). \tag{22}$$

The lower bound gives the earliest possible arrival after release, and the upper bound leaves sufficient time to reach the destination. Nodes violating (22) and their incident arcs are excluded. Shortest-path durations are computed separately for the two modes using their respective eligible links and travel times.

R2: Forward–backward reachability.. The shortest-time bounds are necessary but do not guarantee a feasible path through the candidate network. Let $\mathcal{F}_k^m$ contain the nodes reachable from σ_k^m, and let $\mathcal{B}_k^m$ contain those from which d_k^m is reachable. Retain

$$\mathcal{V}_k^{m,\text{keep}} = \mathcal{F}_k^m \cap \mathcal{B}_k^m \tag{23}$$

and arcs whose endpoints both belong to this set. If the sink is unreachable, remove the corresponding layer network and fix $z_k^{m,\omega} = 0$ for all scenarios.

Both searches take linear time in the network size. They account for the actual candidate arcs, including ROW-block eligibility and the absence of intermediate waiting.

R3: Budget-aware autonomous-path pruning.. For an autonomous travel arc e associated with entry pair (ℓ, t), define its required ROW cells as

$$\mathcal{R}(e) = \{(\ell, p) : p \in \mathcal{P}_{\ell t}^A\}. \tag{24}$$

Set $\mathcal{R}(e) = \varnothing$ for dispatch and delivery arcs. For a source-to-sink autonomous path P, the resource required to activate all its cells is

$$C(P) = \sum_{\xi \in \bigcup_{e \in P} \mathcal{R}(e)} \rho_\xi, \tag{25}$$

where $\xi = (\ell, p)$ and $\rho_\xi = \rho_{\ell p}$.

Different travel arcs on the same path may require the same coarse design cell. Adding its full resource separately for each arc would therefore overestimate $C(P)$. To construct an additive lower bound, let $\bar{n}_{k\xi} \geq 1$ bound the number of arcs of any autonomous path for commodity k that require cell ξ. For each cell appearing in the retained network, a directly available bound is

$$\bar{n}_{k\xi} = \left| \{e \in \mathcal{E}_k^{A,T} : \xi \in \mathcal{R}(e)\} \right|. \tag{26}$$

A tighter bound can be obtained by a longest-path calculation with arc weights $\mathbf{1}\{\xi \in \mathcal{R}(e)\}$.

Assign each autonomous arc the weight

$$\varrho_{ke} = \sum_{\xi \in \mathcal{R}(e)} \frac{\rho_\xi}{\bar{n}_{k\xi}}. \tag{27}$$

Dispatch and delivery arcs have zero weight. For every autonomous path,

$$\sum_{e \in P} \varrho_{ke} \le C(P). \tag{28}$$

Let $P_k^A(a, b)$ denote the directed paths from a to b in the current autonomous network. Define the forward and backward labels

$$\Phi_k^+(v) = \min_{P \in P_k^A(\sigma_k^A, v)} \sum_{e \in P} \varrho_{ke}, \tag{29}$$

$$\Phi_k^-(v) = \min_{P \in P_k^A(v, d_k^A)} \sum_{e \in P} \varrho_{ke}, \tag{30}$$

with value $+\infty$ when the corresponding path set is empty. Delete an autonomous travel arc $e = (u, v)$ if

$$\Phi_k^+(u) + \varrho_{ke} + \Phi_k^-(v) > B. \tag{31}$$

If

$$\Phi_k^+(d_k^A) > B, \tag{32}$$

remove the autonomous network for commodity k and fix $z_k^{A,\omega} = 0$ for all scenarios.

The labels are computed by shortest-path passes on the acyclic network. After pruning, R2 is applied again to remove disconnected nodes and arcs. Further passes of R3 may be performed, but are not required for the validity of the reduction.

Preservation of the original problem.. Let $Q_\omega^R(y)$ denote the recourse value after R1–R3. All subsequent network sets refer to the retained networks, with zero mode allocations imposed whenever a commodity layer has been removed.

Proposition 3 (Exactness of network reduction)**.** *For every binary design $y \in \mathcal{Y}$ and scenario $\omega \in \Omega$,*

$$Q_\omega^R(y) = Q_\omega(y). \tag{33}$$

Consequently, replacing Q_ω by Q_ω^R in (10) *preserves the optimal value and the set of optimal ROW schedules.*

Proof. R1 and R2 remove only nodes and arcs outside temporally feasible source-to-sink paths.

For R3, let $n_{P\xi}$ count the arcs of path P requiring cell ξ. Since $n_{P\xi} \le \bar{n}_{k\xi}$,

$$\sum_{e \in P} \varrho_{ke} = \sum_{\xi \in \bigcup_{e \in P} \mathcal{R}(e)} \rho_\xi \frac{n_{P\xi}}{\bar{n}_{k\xi}} \le C(P).$$

Every path containing $e = (u, v)$ has surrogate weight at least $\Phi_k^+(u) + \varrho_{ke} + \Phi_k^-(v)$. If this quantity exceeds B, the path requires more resources than any feasible binary design can provide.

By Proposition 2, every positive-flow autonomous path under a binary design uses only active ROW cells. Its required resource is therefore at most B. It cannot contain an arc removed by R3. The same argument applies to removal of an entire autonomous layer.

Thus, any feasible recourse solution under a binary design can be represented on the reduced networks without changing its cost. Conversely, every reduced-network solution is feasible on the original networks. The recourse values are equal. □

Budget pruning need not preserve recourse values at fractional designs. The decomposition therefore uses Q_ω^R consistently on $\overline{\mathcal{Y}}$. This reduced recourse function remains finite, continuous, convex, and piecewise linear by the same argument as Proposition 1. Its agreement with Q_ω on $\mathcal{Y}$ is sufficient for exact optimization of the original integer problem.

5.3. Exact Pareto-oriented Benders cuts

Write the reduced scenario problem as

$$Q_\omega^R(y) = \min_{x_\omega \geq 0} q_\omega^\top x_\omega \tag{34}$$

$$\text{s.t.} \quad W_\omega x_\omega \geq h_\omega - T_\omega y, \tag{35}$$

where x_ω collects the retained flow, mode-allocation, and external-service variables. Equalities are represented by pairs of inequalities in this notation; implementations may retain them as equalities with unrestricted dual multipliers.

The dual feasible region is

$$\Pi_\omega = \{\pi_\omega \geq 0 : W_\omega^\top \pi_\omega \leq q_\omega\}, \tag{36}$$

which is independent of y. Strong duality gives

$$Q_\omega^R(y) = \max_{\pi_\omega \in \Pi_\omega} (h_\omega - T_\omega y)^\top \pi_\omega. \tag{37}$$

At a trial schedule $\hat{y}^r$, let $\hat{\pi}_\omega^r$ be an optimal dual solution. The affine function

$$L_\omega^r(y) = (h_\omega - T_\omega y)^\top \hat{\pi}_\omega^r \tag{38}$$

is a lower bound on $Q_\omega^R(y)$ throughout $\overline{\mathcal{Y}}$ and is tight at $\hat{y}^r$. It yields the exact Benders cut

$$\theta_\omega \geq L_\omega^r(y). \tag{39}$$

Alternative dual optima may produce substantially different bounds away from the trial schedule. Maintain a core point $y^{c,r} \in \text{relint}(\overline{\mathcal{Y}})$ and solve

$$\max_{\pi_\omega \in \Pi_\omega} \quad (h_\omega - T_\omega y^{c,r})^\top \pi_\omega \tag{40}$$

$$\text{s.t.} \quad (h_\omega - T_\omega \hat{y}^r)^\top \pi_\omega = Q_\omega^R(\hat{y}^r). \tag{41}$$

The equality restricts the search to the dual optimal face at $\hat{y}^r$. The objective selects the strongest bound at the core point. An optimal solution $\pi_\omega^{P,r}$ replaces $\hat{\pi}_\omega^r$ in (38) to generate the Pareto-oriented cut.

For $B > 0$ and $\mathcal{C} \neq \varnothing$, initialize

$$y_{\ell p}^{c,0} = \eta, \qquad 0 < \eta < \min\left\{1, \frac{B}{\sum_{(\ell,p)\in\mathcal{C}} \rho_{\ell p}}\right\}. \tag{42}$$

Update the core point by

$$y^{c,r+1} = \gamma y^{c,r} + (1-\gamma)\hat{y}^r, \qquad 0 < \gamma < 1. \tag{43}$$

The base implementation uses $\gamma = 0.7$. If $B = 0$ or $\mathcal{C} = \varnothing$, the only design is $y = 0$, and the scenario problems are solved directly.

The coefficient of $y_{\ell p}$ has a useful capacity interpretation. Let $\mu_{\ell t p}^{\omega r} \geq 0$ multiply the autonomous constraint written as $F_{\ell t}^{A,\omega} - U_{\ell t}^{A,\omega} y_{\ell p} \leq 0$ in the minimization Lagrangian. Similarly, let $\nu_{\ell t}^{\omega r} \geq 0$ multiply the conventional capacity constraint with its right-hand side moved to the left. Then

$$\beta_{\ell p}^{\omega r} = -\sum_{\substack{t:(\ell,t)\in\mathcal{A}^A \\ p\in\mathcal{P}_{\ell t}^A}} U_{\ell t}^{A,\omega} \mu_{\ell t p}^{\omega r} + \sum_{\substack{t\in\mathcal{T}_p \\ (\ell,t)\in\mathcal{A}^H}} \Delta_{\ell t} \nu_{\ell t}^{\omega r}. \tag{44}$$

The first term aggregates autonomous traversals requiring block p. The second aggregates conventional capacity removed during that block. These are the only sources of dependence on y in the scenario formulation.

5.4. Commodity-separable network-relaxation cuts

The exact scenario problem couples commodities through shared capacities. We obtain a lower bound by assigning each commodity the full capacity of every link–period pair, while retaining its timing, routing, and mode-allocation constraints.

For commodity k, define

$$\underline{Q}^R_{k\omega}(y) = \min_{f_k, z_k, s_k} \quad \sum_{m\in\mathcal{M}} \sum_{e\in\mathcal{E}^m_k} c^{m,\omega}_e f^{m,\omega}_{ke} + M_k s^\omega_k \tag{45}$$

$$\text{s.t.} \quad z^{A,\omega}_k + z^{H,\omega}_k + s^\omega_k = d^\omega_k, \tag{46}$$

$$\sum_{e\in\delta^+_{km}(v)} f^{m,\omega}_{ke} - \sum_{e\in\delta^-_{km}(v)} f^{m,\omega}_{ke} = b^m_{kv} z^{m,\omega}_k, \qquad m\in\mathcal{M},\ v\in\mathcal{V}^m_k, \tag{47}$$

$$\sum_{e\in\mathcal{E}^{A,T}_{k,\ell t}} f^{A,\omega}_{ke} \le U^{A,\omega}_{\ell t} y_{\ell p}, \qquad \begin{array}{l}(\ell,t)\in\mathcal{A}^A,\\ p\in\mathcal{P}^A_{\ell t},\end{array} \tag{48}$$

$$\sum_{e\in\mathcal{E}^{H,T}_{k,\ell t}} f^{H,\omega}_{ke} \le R^{H,\omega}_{\ell t} - \Delta_{\ell t} y_{\ell,\kappa(t)}, \qquad \begin{array}{l}(\ell,t)\in\mathcal{A}^H,\\ (\ell,\kappa(t))\in\mathcal{C},\end{array} \tag{49}$$

$$\sum_{e\in\mathcal{E}^{H,T}_{k,\ell t}} f^{H,\omega}_{ke} \le R^{H,\omega}_{\ell t}, \qquad \begin{array}{l}(\ell,t)\in\mathcal{A}^H,\\ (\ell,\kappa(t))\notin\mathcal{C},\end{array} \tag{50}$$

$$f^{m,\omega}_{ke} \ge 0, \qquad m\in\mathcal{M},\ e\in\mathcal{E}^m_k, \tag{51}$$

$$z^{m,\omega}_k \ge 0, \qquad s^\omega_k \ge 0, \qquad m\in\mathcal{M}. \tag{52}$$

Empty arc sums are zero. If preprocessing removes a commodity layer, its mode allocation remains fixed at zero. The demand equality and nonnegativity already imply $z^{m,\omega}_k \le d^\omega_k$.

Define the separable scenario value

$$\underline{Q}^R_\omega(y) = \sum_{k\in\mathcal{K}} \underline{Q}^R_{k\omega}(y). \tag{53}$$

Each commodity problem has a dual feasible region independent of y. Let $\Pi^N_{k\omega}$ denote this region and write its dual objective as $(h^N_{k\omega} - T^N_{k\omega} y)^\top \pi^N_{k\omega}$.

At reference schedule $\bar{y}^j \in \overline{\mathcal{Y}}$, obtain an optimal dual solution $\bar{\pi}^{N,j}_{k\omega}$ and define

$$\alpha^j_{k\omega} = (h^N_{k\omega})^\top \bar{\pi}^{N,j}_{k\omega}, \tag{54}$$

$$\beta^j_{k\omega} = -(T^N_{k\omega})^\top \bar{\pi}^{N,j}_{k\omega}. \tag{55}$$

Summing these bounds gives

$$N^j_\omega(y) = \alpha^j_\omega + \sum_{(\ell,p)\in\mathcal{C}} \beta^{\omega j}_{\ell p} y_{\ell p}, \tag{56}$$

where $\alpha^j_\omega = \sum_k \alpha^j_{k\omega}$ and $\beta^{\omega j}_{\ell p} = \sum_k (\beta^j_{k\omega})_{\ell p}$. The resulting master cut is

$$\theta_\omega \ge N^j_\omega(y). \tag{57}$$

Its capacity coefficients follow (44), summed over the commodity-specific multipliers.

Proposition 4 (Validity of the network-relaxation cuts)**.** *For every $y\in\overline{\mathcal{Y}}$,*

$$N^j_\omega(y) \le \underline{Q}^R_\omega(y) \le Q^R_\omega(y). \tag{58}$$

At the reference schedule, $N^j_\omega(\bar{y}^j) = \underline{Q}^R_\omega(\bar{y}^j)$.

Proof. Take a feasible solution of the joint reduced scenario problem. Nonnegative commodity flows imply that the use of any capacity row by one commodity is no greater than its aggregate use. The projected solution for commodity k is therefore feasible for its relaxation. Summing the commodity optimal values gives $\underline{Q}^R_\omega(y) \le Q^R_\omega(y)$.

Each affine bound is generated by a feasible commodity dual solution. Weak duality and summation give $N^j_\omega(y) \le \underline{Q}^R_\omega(y)$. Optimality of these dual solutions at $\bar{y}^j$ gives equality with the separable value there. □

These cuts are tight for the relaxation at their reference schedules, but may underestimate the joint recourse value because they omit competition for capacity. They supply initial lower bounds and do not replace exact scenario evaluation.

The base implementation generates them only during initialization. For each scenario, solve the commodity relaxations at $y = 0$ and at the initial core point. After adding these cuts, solve the preliminary master to obtain $(\hat{y}^N, \hat{\theta}^N)$. Generate one additional cut for scenario ω only if

$$\underline{Q}^R_\omega(\hat{y}^N) - \hat{\theta}^N_\omega > \epsilon_N. \tag{59}$$

Duplicate reference schedules are skipped. At most three network-relaxation cuts are therefore generated per scenario.

For each scenario and reference schedule, the commodity problems may be assembled into one block-diagonal LP. Its objective equals $\underline{Q}^R_\omega$, and its block duals produce the same aggregated cut with fewer solver calls.

5.5. Parallel multi-cut Benders algorithm

Let B_ω index the accepted exact cuts and N_ω the initialization cuts from the network relaxation. The restricted master is

$$\min_{y,\theta} \quad \sum_{(\ell,p)\in\mathcal{C}} g_{\ell p} y_{\ell p} + \sum_{\omega\in\Omega} p_\omega \theta_\omega \tag{60}$$

$$\text{s.t.} \quad y \in \mathcal{Y}, \tag{61}$$

$$\theta_\omega \ge 0, \qquad \omega \in \Omega, \tag{62}$$

$$\theta_\omega \ge L^r_\omega(y), \qquad \omega \in \Omega,\ r \in B_\omega, \tag{63}$$

$$\theta_\omega \ge N^j_\omega(y), \qquad \omega \in \Omega,\ j \in N_\omega. \tag{64}$$

At iteration r, let $(\hat{y}^r, \hat{\theta}^r)$ be a feasible integer master solution. Let b^r_{MP} be the solver's valid lower bound on the master optimum. Update

$$LB_r = \max\{LB_{r-1}, b^r_{\mathrm{MP}}\}. \tag{65}$$

If the master is solved to optimality, b^r_{MP} equals its objective value. Otherwise, the solver's best bound is used, rather than its incumbent objective.

Solve the scenario problems at $\hat{y}^r$. The evaluated design yields the upper-bound update

$$UB_r = \min\left\{ UB_{r-1},\ \sum_{(\ell,p)\in\mathcal{C}} g_{\ell p}\hat{y}^r_{\ell p} + \sum_{\omega\in\Omega} p_\omega Q^R_\omega(\hat{y}^r) \right\}. \tag{66}$$

The corresponding design and scenario flows are stored whenever the incumbent improves. The stopping criterion is

$$\frac{UB_r - LB_r}{\max\{1, |UB_r|\}} \leq \epsilon. \tag{67}$$

Algorithm: Parallel multi-cut Benders with Pareto and network-relaxation cuts

1. Apply R1–R3 and construct the reduced scenario networks, retaining the original candidate set $\mathcal{C}$. If $B = 0$ or $\mathcal{C} = \varnothing$, solve all scenarios at $y = 0$ and return the resulting solution.
2. Initialize $y^{c,0}$ using (42). Generate network-relaxation cuts at 0 and $y^{c,0}$, solve the preliminary master, and generate the optional third cut according to (59).
3. Set $LB = 0$, $UB = +\infty$, and $r = 0$.
4. Solve the restricted master to obtain $(\hat{y}^r, \hat{\theta}^r)$. Update LB using (65).
5. In parallel over scenarios, solve $Q_\omega^R(\hat{y}^r)$ and retain optimal dual solutions. Update the incumbent and UB using (66). If (67) holds, return the incumbent and final bounds.
6. For each scenario satisfying $Q_\omega^R(\hat{y}^r) - \hat{\theta}_\omega^r > \epsilon_{\text{cut}}$, solve (40)–(41). Add the resulting exact cut if the dual-feasibility and tightness checks pass. Otherwise, use the optimal dual solution obtained from the recourse solve.
7. Update the core point using (43), set $r \leftarrow r + 1$, and return to Step 4. If the gap exceeds tolerance but no cut is accepted, tighten the separation tolerance and recheck the underestimated scenarios.

The implementation uses a consistent dual-sign convention across all scenario models. Exact cuts are checked for dual feasibility and tightness against Q_ω^R at their generation points. Network-relaxation cuts are checked against $\underline{Q}_\omega^R$ at their reference points. The ordinary optimal-dual fallback permits the Benders iteration to continue when the Pareto auxiliary problem encounters numerical difficulties.

Scenario problems are solved in parallel under a fixed total thread budget. All accepted exact cuts are retained, and the network-relaxation family is restricted to initialization. Reported computation times include preprocessing, initialization, master optimization, and scenario solves.

For a finite scenario set, exact master and scenario solutions, and zero stopping and separation tolerances, the algorithm terminates finitely with an optimal solution. Every underestimated binary design generates an exact cut that is tight at that design. It cannot remain underestimated after that cut is added. Since $\mathcal{Y}$ contains finitely many binary designs and all initialization cuts are valid, the standard finite-convergence argument applies. Proposition 3 transfers the resulting optimality guarantee to the original problem.

6. Computational study

The experiments evaluate computational performance and the operating value of temporary nighttime ROW allocation. R04–R10 provide the algorithmic test set. A Sioux Falls instance with 30 selected OD pairs supports policy comparisons and spatial analysis. All networks use the same temporal resolution, but their capacity and cost calibrations differ; policy results are therefore reported separately from the R benchmark comparisons.

6.1. Instances and experimental settings

The planning horizon is 00:00–06:00. First-stage decisions use six one-hour design blocks, while second-stage movements use 36 ten-minute periods. Thus, $\mathcal{T} = \{0, \ldots, 36\}$, $\mathcal{P} = \{1, \ldots, 6\}$, and $\kappa(t) = 1 + \lfloor t/6 \rfloor$ for $t = 0, \ldots, 35$. All directed links are ROW candidates in every design block. Table 2 reports the dimensions before preprocessing.

Table 2: Benchmark dimensions under the one-hour design grid.

Network	Nodes	Links	Commodities	ROW binaries
R04	10	60	10	360
R05	10	60	25	360
R06	10	60	50	360
R07	10	82	10	492
R08	10	83	25	498
R09	10	83	50	498
R10	20	120	40	720
Sioux Falls 30OD	24	76	30	456

Each physical OD defines one commodity with a single release time and deadline. ODs are not replicated across release waves. The original OD pairs, mean demands, release times, and 30 demand realizations are retained from the previous data package. Link travel times are converted through their nominal durations in minutes, rather than by reusing old period counts. Deadlines are rounded inward to the new grid. Appendix **??** gives the conversion rules.

Each workbook contains three cost–capacity regimes, $Y \in \{1, 3, 5\}$. In the R files, Y1 has lower fixed charges, Y3 has higher fixed charges with the same capacities, and Y5 has intermediate fixed charges and generally tighter capacities. These labels do not denote demand multipliers. Sioux Falls uses calibrated regimes with the same ordering. The scenario subsets contain records 1–10, 1–20, and 1–30, with probabilities $1/S$ within a selected sample of size S. Demand and traffic shocks are shared across the three regimes.

The baseline ROW share is 20% of link capacity, and autonomous entry capacity is 1.25 times the transferred capacity. The deployment budget is 25% of the full six-hour resource total. Splitting the former two-hour blocks into hours preserves the total resource and full-night activation cost of each link.

All algorithm variants use the same input records, solver tolerances, total thread budget, and time limit. We prescribe a 3,600-second wall-clock limit and a relative optimality tolerance of 10^{-4} for the main comparisons. Hardware, solver version, thread allocation, and

numerical tolerances are recorded with every run. Wall-clock time includes preprocessing, NR initialization, Pareto auxiliary solves, and the main decomposition loop.

6.2. Algorithm correctness and computational performance

6.2.1. Correctness verification

Small instances are solved both as deterministic-equivalent MILPs and by Benders decomposition. A reproducible reduced R04 test retains its physical network, the first two commodity records, and the first two scenarios, with scenario probabilities renormalized to one half. At most 12 candidate cells are allowed to open. These cells are selected from autonomous shortest-path traversals departing at the two release times, sorted by link and block index; if more than 12 are required, retain the first 12. All other design cells are fixed to zero. The restricted candidate set is identical in both methods. External service preserves feasibility even when no complete ROW path remains.

We compare optimal objective values, valid lower and upper bounds, and primal constraint residuals. Multiple optimal designs need not have identical deployment costs or activation patterns. Each returned design is also evaluated by an independently constructed recourse LP. For selected binary designs, recourse values before and after R1–R3 must agree. Fractional cut checks use the same reduced recourse formulation as the decomposition. For exact cuts, we verify dual feasibility and tightness at the generation point; NR cuts are checked for tightness against the separable relaxation and for lying below the joint reduced recourse value.

Table 3: Correctness verification. Blank cells await completed runs.

Test	DE objective	BD objective	Relative difference	Max. residual
Small R04, 2 scenarios				
Small R04, 1 scenario				

6.2.2. Contribution of the cuts

We compare BD, BD+P, and BD+P+NR on R04–R10 with Y1 and ten scenarios, giving 21 runs. BD uses ordinary optimal-dual cuts. BD+P replaces their selection by the Pareto procedure. BD+P+NR additionally uses the root NR cuts. All three variants use R1–R3 and the same initial bound $\theta_\omega \geq 0$. This comparison measures the contribution of Pareto selection and the incremental contribution of NR after Pareto selection; it does not isolate the performance of NR used alone.

Table 4 reports wall-clock time, final gap, and exact Benders iteration count. Initialization solves are included in time but not in the iteration count. An improvement is useful only if its reduction in iterations or gap compensates for its additional computation. For two representative networks, plot relative gap against elapsed time to show whether an enhancement improves early bounds, later convergence, or both.

6.2.3. Scalability across benchmark regimes

BD+P+NR is evaluated on all $7 \times 3 \times 3 = 63$ combinations of network, Y regime, and sample size. The seven overlapping runs from the cut comparison may be reused when all

Table 4: Cut comparison on Y1 with ten scenarios. Time is in seconds; gap is in percent.

	BD			BD+P			BD+P+NR		
Network	Time	Gap	Iter.	Time	Gap	Iter.	Time	Gap	Iter.
R04									
R05									
R06									
R07									
R08									
R09									
R10									

settings are identical. For each case, record time, final relative gap, lower bound, upper bound, and termination status. The principal table reports time, gap, and status; objective bounds remain available in the accompanying machine-readable results. The reported percentage gap is

$$100\frac{UB - LB}{\max\{1, |UB|\}}.$$

For a time-limited master, LB uses the solver's valid best bound, not the master incumbent objective. "Solved" means that the prescribed gap tolerance has been reached; it does not mean that an exact zero gap has been proved.

The analysis separates the effects of commodity count, network density, cost–capacity regime, and scenario count. R04–R10 are not ordered by a single size measure. Summaries report solved counts and remaining gaps for unsolved cases, rather than treating every time-limit observation as a solution time.

Table 5: Full benchmark results for BD+P+NR.

Network	Y	S	Time (s)	Gap (%)	Status	Notes
R04	1	10				
R04	1	20				
R04	1	30				
R04	3	10				
R04	3	20				
R04	3	30				
R04	5	10				
R04	5	20				
R04	5	30				
R05	1	10				
R05	1	20				
R05	1	30				
R05	3	10				
R05	3	20				
R05	3	30				

Network	Y	S	Time (s)	Gap (%)	Status	Notes
R05	5	10				
R05	5	20				
R05	5	30				
R06	1	10				
R06	1	20				
R06	1	30				
R06	3	10				
R06	3	20				
R06	3	30				
R06	5	10				
R06	5	20				
R06	5	30				
R07	1	10				
R07	1	20				
R07	1	30				
R07	3	10				
R07	3	20				
R07	3	30				
R07	5	10				
R07	5	20				
R07	5	30				
R08	1	10				
R08	1	20				
R08	1	30				
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R08	5	20				
R08	5	30				
R09	1	10				
R09	1	20				
R09	1	30				
R09	3	10				
R09	3	20				
R09	3	30				
R09	5	10				
R09	5	20				
R09	5	30				
R10	1	10				
R10	1	20				
R10	1	30				
R10	3	10				

Network	Y	S	Time (s)	Gap (%)	Status	Notes
R10	3	20				
R10	3	30				
R10	5	10				
R10	5	20				
R10	5	30				

6.2.4. Network reduction

On R04–R10, Y1, and ten scenarios, record network size after no reduction, R1, R1–R2, and R1–R3. Counts are summed over commodity-specific A and H networks before scenario replication. The number of flow variables in one scenario equals the retained commodity-arc count. Report preprocessing time and the final arc reduction $100(1 - |\mathcal{E}^{\text{final}}|/|\mathcal{E}^{\text{raw}}|)$. Here the raw network includes structurally eligible arcs before commodity time-window screening. Candidate-eligibility checks are part of construction, not an additional reduction rule.

First-stage binaries are unchanged because R4 is not used. No full optimization run is required for every intermediate reduction stage. If R3 deletes few or no arcs at the baseline budget, report that outcome. An optional tighter-budget diagnostic can examine when its resource bound becomes useful, but is reported separately from the baseline experiment.

Table 6: Sequential network reduction. Counts include both modes and all commodities.

Network	Stage	Nodes	Arcs	Time (s)	Reduction (%)
R04	None				
R04	R1				
R04	R1–R2				
R04	R1–R3				
R05	None				
R05	R1				
R05	R1–R2				
R05	R1–R3				
R06	None				
R06	R1				
R06	R1–R2				
R06	R1–R3				
R07	None				
R07	R1				
R07	R1–R2				
R07	R1–R3				
R08	None				
R08	R1				
R08	R1–R2				
R08	R1–R3				

Network	Stage	Nodes	Arcs	Time (s)	Reduction (%)
R09	None				
R09	R1				
R09	R1–R2				
R09	R1–R3				
R10	None				
R10	R1				
R10	R1–R2				
R10	R1–R3				

6.3. Value of stochastic and time-dependent ROW allocation

The Sioux Falls case retains all 24 nodes and 76 directed links but only 30 selected OD commodities. We compare four policies under each Y regime: no ROW, an expected-value design (EV), a static full-night ROW design, and the stochastic hourly design. EV replaces uncertain demand and residual capacities by their training-sample means, optimizes once, and evaluates the resulting design under uncertainty. Means of residual capacity are computed after clipping background shares. Static ROW imposes $y_{\ell 1} = \cdots = y_{\ell 6}$ and uses the same total budget and the sum of the hourly deployment costs. Both static and hourly stochastic designs use the same 30 training scenarios.

All fixed designs are evaluated on the same independent 1,000-scenario sample. For policy a, define

$$\widehat{J}_a = \sum_{\ell,p} g_{\ell p} y^a_{\ell p} + \frac{1}{N} \sum_{\omega=1}^{N} Q_\omega(y^a), \qquad N = 1000.$$

Use the same original or temporally screened evaluation networks for every policy. Budget pruning must be recomputed if the budget changes; a network pruned for a smaller budget cannot be reused for a larger one. We report expected total cost, savings relative to no ROW, autonomous and external-service shares, and resource-budget utilization. Mode shares divide total assigned volume across evaluation scenarios by total realized demand, rather than averaging unweighted scenario ratios.

Paired scenario cost differences support confidence intervals for policy comparisons. Deployment-cost differences enter every paired observation. For baseline-minus-policy cost differences Δ_ω, report the approximate 95% interval $\bar{\Delta} \pm 1.96 s_\Delta / \sqrt{N}$ in cost units, conditional on the trained designs. Percentage savings are reported separately as point estimates. The EV comparison estimates the benefit of the computed stochastic design; sampling and optimization error may prevent a positive observed difference. No sign is imposed on the reported result.

The spatial presentation uses six panels, one for each hourly design block, with selected ROW links highlighted and autonomous entry flows shown by line width. A second plot reports A, H, and external volumes by commodity release hour. These views reveal whether openings form usable routes, shift with release times, and concentrate on shared corridors. They do not presume that the optimal solution exhausts the budget or forms one connected network.

Table 7: Out-of-sample policy comparison on Sioux Falls.

Y	Policy	Total cost	Savings (%)	A share (%)	External (%)	Budget (%)
1	No ROW					
1	EV					
1	Static ROW					
1	Stochastic ROW					
3	No ROW					
3	EV					
3	Static ROW					
3	Stochastic ROW					
5	No ROW					
5	EV					
5	Static ROW					
5	Stochastic ROW					

6.4. Resource and operating-condition sensitivity

We vary one factor at a time on Sioux Falls Y1 and Y5, retaining the same underlying random draws. Table 8 defines the levels. Each design is reoptimized and evaluated under its corresponding condition. The baseline point can be reused across sweeps. Y3 remains part of the policy comparison and can be added to a sweep if its higher activation costs require further explanation.

Table 8: One-factor sensitivity experiments.

Factor	Levels	Transformation
Budget share	$0, 0.125, 0.25, 0.375, 0.50$	Change B as a share of the full resource total.
Background multiplier	$0.5, 1, 1.5, 2$	Multiply the deterministic hourly background profile before adding the stored shocks and clipping.
Demand multiplier	$0.75, 1, 1.25, 1.5$	Multiply every mean and realized demand by the same factor.
Autonomous cost multiplier	$0.8, 1, 1.2$	Multiply autonomous travel-arc costs; keep capacity and travel time fixed.

For each sweep, show total cost and mode shares, with budget utilization or active link–block counts as supporting measures. Larger budgets weakly improve the optimal in-sample objective because the feasible set expands, but sampled out-of-sample costs and computed solutions need not be monotone. Temporal connectivity may produce thresholds at which an additional set of openings first supports a useful route. Their occurrence and location are empirical questions, not assumed conclusions.

To assess the capacity trade-off, report conventional rerouting and the share of reserved link–period observations whose conventional capacity is binding. Only positive-load observations are used for utilization ratios. A limited diagnostic sets $\Delta = 0$ while keeping

autonomous capacity at its baseline value. This is a counterfactual without conventional capacity loss, not a recalibration in which U^A is also set to zero. Its difference from the baseline helps establish whether capacity competition materially affects ROW decisions. If the effect is small under light traffic, the discussion should emphasize route timing and resource allocation instead.

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